Forecast, Flag, Shed: A Learned Emergency Controller for Loss-of-Load

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Abstract

Learned components are usually validated on aggregate accuracy in isolation, but the systems they run inside fail on rare events and on operating regimes the training sample never saw. We measure that gap on a concrete safety-critical system, an operator who must shed load when generation falls short, and we score the learned emergency controller by the closed-loop outcome rather than by standalone accuracy. Three questions organize the evaluation. Can a forecaster predict rare failures ahead of time? At the 24-hour lead it reaches AUROC 0.963 with expected calibration error 0.0005, and it survives a structural shift to a renewables-heavy fleet at AUROC 0.986 while the classical capacity-outage baseline stays at 0.889. Can a learned policy act on that risk under tight frequency constraints? It blackouts 38% of a 200-scenario stress set, against 45% for a fixed oracle and 81% for no action. Does the pair prevent blackouts on the hours it chooses? On the 80 flagged hours it holds blackouts to 3%, against 36% for staged relays and 64% for no action, while shedding no more than the oracle. Every number regenerates from saved artifacts. The result that generalizes past this grid is that a component's accuracy and its contribution inside a loop are different quantities, and only the second one decides whether the system is safe.

1 Introduction

A loss-of-load event happens when customer demand exceeds the generation that is actually available, and the quantity an operator needs is its probability before the event rather than its frequency after it. As inverter-based resources displace synchronous generation the margin between load and available capacity thins and the operating regime shifts structurally from one year to the next. The standard indices, loss-of-load probability LOLP and loss-of-load expectation LOLE, are computed offline, either by sequential Monte Carlo over years of synthetic operation or by a recursive convolution that builds a capacity-outage table under the assumption that unit failures are independent [Billinton and Allan, 1996]. Both are correct as long-run statistics, both are silent about tomorrow, and neither speaks to the control loop that has to act when the deficit arrives.

The learning-based literature has attacked the pieces separately. One thread predicts adequacy indices directly from system features and replaces slow Monte Carlo with end-to-end models [Zhu and Singh, 2022, Sharifnia and Tindemans, 2022], while another trains reinforcement-learning agents to shed load during frequency excursions because the fixed relay stages of classical under-frequency load shedding (UFLS) are too rigid to time their sheds to the disturbance [Justin and Paternain, 2024, Chen et al., 2023]. Neither closes the loop between them, since a forecast of loss-of-load risk is not a decision and a shed policy that fires only after the frequency has collapsed has thrown away the hours of warning the forecast could have supplied.

This paper builds that connection, and the contribution is a method rather than an observation. A pipeline *forecasts* a calibrated loss-of-load probability 1 to 24 hours ahead from information observable at hour t, *flags* the hours worth acting on with an isotonic-calibrated threshold, and *sheds*

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with a deep Q-network trained on a reduced-order frequency-dynamics model when a contingency hits a flagged hour. Two measured results carry it. First, the forecaster beats the exact classical baseline at the day-ahead lead, AUROC 0.963 against 0.890, because it captures the serial dependence of unit outages that the independence assumption throws away, and its calibration survives a structural shift to a renewables-era mix at AUROC 0.986. Second, the closed loop works, because on the 80 hours the flagger selected over five test years, 0.18% of hours at 80% precision, the learned shed policy holds blackouts to 0.03 against 0.64 with no action and 0.36 with staged UFLS. The paper also ships a compact seeded simulator, matched-scenario comparisons, per-seed records, and a machine-checkable reproduction trail.

2 Problem formulation

Generation-adequacy simulator. We simulate an RTS-inspired fleet of $n = 10$ thermal units with capacities 20 to 197 MW and forced outage rates 0.02 to 0.10 [Billinton and Allan, 1996]. Hourly load L_t follows a weather-driven model with annual, weekly, and daily components and an AR(1) temperature process, and the seasonal peak is 800 MW against 784 MW of installed capacity, so loss of load occurs at the roughly 9 h/yr scale typical of a tight system. Ground truth is sequential Monte Carlo, in which each unit alternates up and down with exponential sojourn times, mean time to failure follows from the forced outage rate, and hour t sees a deficit $\Delta_t = \max(0,\, L_t - A_t)$ when the available capacity A_t falls below L_t. The per-hour index is $\mathrm{LOLP}_t = \mathbb{P}(\Delta_t > 0)$, LOLE sums LOLP over a horizon in h/yr, and EENS is the expected deficit in MWh/yr. The classical baseline is the capacity outage probability table (COPT), an exact recursive convolution over the marginal outage rates that gives LOLP for a known load under the independence assumption [Billinton and Allan, 1996].

Frequency-dynamics environment. For the control problem we use a reduced-order single-area frequency model, which is the standard adequacy-scale abstraction [Anderson and Mirheydar, 1992] and not an AC power-flow model. In per-unit system base,

$$\frac{d\Delta f}{dt} = \frac{P_{\mathrm{shed}} + P_g - P_{\mathrm{lost}} - D\,\Delta f}{2H}, \qquad \frac{dP_g}{dt} = \frac{\min(-\Delta f/R,\; P_{\mathrm{res}}) - P_g}{T_g}, \tag{1}$$

where Δf is the frequency deviation from 60 Hz nominal, $H = 4\,\mathrm{s}$ is inertia, $D = 1.2$ is load damping, $R = 0.05$ is droop, $T_g = 3\,\mathrm{s}$ is the governor time constant, P_{lost} is the tripped generation, P_{res} is the spinning reserve, and P_{shed} is the shed load, all in per-unit on the system base except H, R, and T_g. A blackout is declared when frequency falls below 57.5 Hz and recovery when it sustains above 59.7 Hz, and staged UFLS relays latch 5% sheds at 59.3, 58.9, 58.7, and 58.5 Hz, matching common practice [Anderson and Mirheydar, 1992]. The environment is vectorized over parallel episodes for batched training, and it is deliberately single-area, which Section 5 states as a limit.

Learning problems. The forecasting problem takes the features observed at hour t, which are the forecast load and temperature at $t + \ell$, calendar terms, current available capacity and reserve, and the unit-outage history including a causal 24-hour rolling loss-of-load rate, and asks for $y_t = \mathbb{1}[\Delta_{t+\ell} > 0]$, the indicator of a deficit at lead $\ell \in \{1, 24\}$ h. The control problem is a Markov decision process with state $(\Delta f,\, P_g,\, P_{\mathrm{shed}},\, P_{\mathrm{lost}},\, P_{\mathrm{res}},\, t)$ and an action drawn from a ten-level set of shed fractions, rewarded by a sum of $-150 \cdot \mathbb{1}[\mathrm{blackout}]$, $+5 \cdot \mathbb{1}[\mathrm{recover}]$, $-10 \cdot E_{\mathrm{shed}}$ and small penalties on nadir depth and elapsed time.

3 Method

Forecaster and flagger. An XGBoost classifier with 400 trees of depth 6 at learning rate 0.05 is trained on 20 years of simulator history, validated on 5 years, and tested on the final 5, with the same split used for both leads within a seed. Predictions are calibrated by isotonic regression fit on the validation set, which turns raw scores into probabilities an operator can threshold. The baselines are the COPT table given the true target-hour load, which is a perfect-load classical forecast, and persistence, the rolling loss-of-load rate over the preceding 24 h, with a random forest as a second learned model. Five seeds re-realize the load process and the outage history, and each test set holds about 43,800 hours. The flagger is the calibrated forecaster thresholded at 0.1, and it chooses the hours the controller prepares for.

Shed policy. A deep Q-network with two hidden layers, experience replay, a target network, and ε-greedy exploration [Mnih et al., 2015] is trained for 24,000 batched episodes per seed on

Table 1: Forecasting loss of load at leads 1 and 24 h, mean ± std over 5 seeds with $n \approx 43{,}800$ test hours per seed. Baselines are the COPT recursive convolution given the true target-hour load and 24-hour persistence, and the last column evaluates the same trained models out of distribution on a structurally different renewables-era fleet.

	lead 1 h		lead 24 h			24 h out of dist.
Model	AUROC	Brier	AUROC	Brier	ECE	AUROC
XGBoost (ours)	0.999±0.001	0.0004	0.963±0.017	0.0010	0.0005	0.986±0.006
Random forest	0.985±0.016	0.0005	0.928±0.049	0.0010	0.0004	0.950±0.037
COPT	0.943±0.013	0.0010	0.890±0.018	0.0010	0.0003	0.889±0.021
Persistence	0.892±0.028	0.0009	0.615±0.057	0.0013	0.0014	0.584±0.102

contingencies sampled as one to four capacity-weighted unit failures against loads drawn from the upper load distribution, so the training distribution is the near-peak hours where a deficit is possible. Three baselines share the same matched scenario set of 200 contingencies, namely no action, staged UFLS, and an oracle that sheds the full deficit immediately.

The loop. The calibrated forecaster flags hour t when its probability exceeds 0.1, and at each flagged hour a contingency is sampled and the no-action, staged, and learned policies are rolled out on the *same* scenario, so the comparison is matched rather than averaged. The system-level measure is the blackout rate across flagged hours under each policy.

4 Results

Result 1: calibrated day-ahead loss-of-load forecasting. The first question is whether loss of load is predictable ahead of time at all, and by which mechanism. The classical answer says the best a forecaster can do is combine a load forecast with the marginal outage statistics, which is what the COPT table computes, while the learned answer says the serial state of the fleet carries information the marginals do not. Figure 1a measures the difference and Table 1 gives the numbers. At the 1-hour lead the learned models are nearly perfect, at AUROC 0.999 and Brier 0.0004, because the availability of the fleet at t almost determines the outcome at $t + 1$, and COPT reaches only 0.943 because it is blind to the current outage state. At the day-ahead lead the gap widens, with XGBoost holding AUROC 0.963 at Brier 0.0010 and expected calibration error 0.0005 while COPT falls to 0.890 and persistence collapses to 0.615. The mechanism is serial dependence. A unit that is down at t is likely still down at $t+24$ because mean time to repair is 40 to 60 hours, so the outage history observed at t is genuinely predictive and the independence assumption discards exactly that information. Calibration separates from discrimination here, since Figure 1b shows the learned and the classical probabilities both tracking observed frequency in aggregate at expected calibration error 0.0005 and 0.0003, so the gap between them is discrimination rather than calibration, because the learned model concentrates probability on the hours that actually fail, which is the property a flagger needs.
The structural-shift test asks whether the model survives a grid it was not trained on. Trained on the thermal fleet and evaluated on a renewables-era fleet of 13 smaller units with higher effective outage rates at comparable total capacity, the learned forecaster's day-ahead AUROC *rises* to 0.986±0.006 at expected calibration error 0.0004, because its features carry the fleet state directly, while the COPT baseline stays at 0.889 and persistence at 0.584.

Result 2: closing the loop converts forecast risk into blackout avoidance. A calibrated risk signal is not yet a decision, and here a component metric stops being enough, because Result 1 scored a forecaster that cannot shed load and its score says nothing about whether acting on it helps. We therefore score the system, first on matched scenarios and then on the hours the forecaster actually flags. On the matched 200-scenario stress set, Figure 2a, no action blackouts 81% of scenarios, staged UFLS 62.5%, and the fixed oracle that sheds the whole deficit immediately 45%. The learned policy reaches a blackout rate of 0.382±0.031, better than the oracle itself, with the highest frequency nadir at 58.4 Hz against 57.9 and the fastest recovery at 2.6 s against 4.1 s, all at slightly less shed energy, 0.16 against 0.18 MWh. It beats the oracle because it times sheds to the state, shedding early and hard only when the deficit demands it instead of dumping the full deficit at $t = 0$. Training takes about 2.5 minutes per seed and the per-seed blackout rates are tight, at 0.355, 0.425, and 0.365.
The loop then measures the system-level effect, which is what happens when the forecaster decides *which* hours to prepare for. Over the five test years the calibrated flagger at threshold 0.1 selected 80 hours, 0.18% of all hours, at precision 0.80, so a contingency is present at a flagged hour four times in

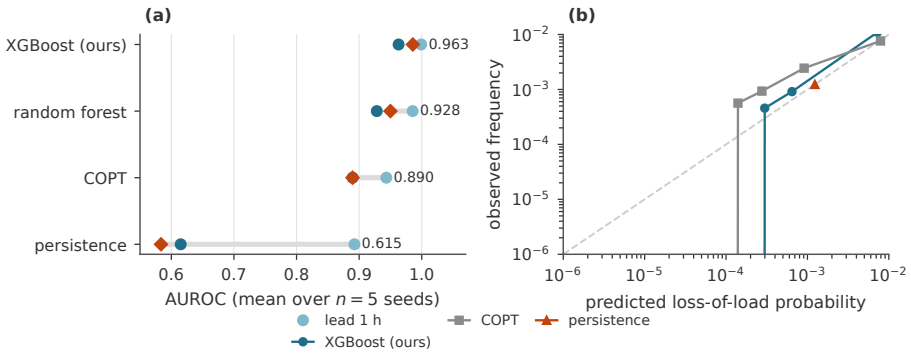


Figure 1: Forecast quality over $n = 5$ seeds at about 43,800 test hours per seed, with the COPT recursive convolution given the true target-hour load and 24-hour rolling persistence as baselines. Panel (a) is AUROC per model at the 1-hour and 24-hour leads and at the 24-hour lead on the structurally different renewables-era fleet, with the 24-hour values printed; panel (b) is the reliability diagram at the 24-hour lead for seed 0, quantile-binned on log axes.

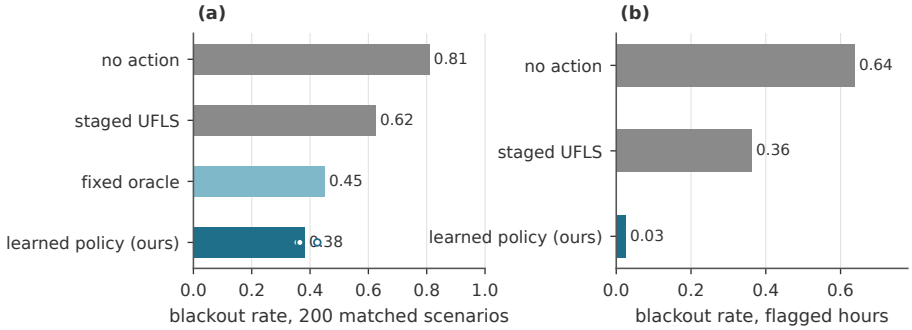


Figure 2: Blackout rate by policy on the same scenarios for every policy. Panel (a) is the offline matched 200-scenario stress set, where the learned policy is its mean over 3 training seeds with the seeds overlaid as open circles against single-run baselines; panel (b) is the closed loop at the 80 hours flagged over 5 test years at threshold 0.1 and flag precision 0.80.

five and the flagger is selective rather than alarmist. Figure 2b rolls all three policies out on the same scenarios at those hours, where no action blackouts 0.64 of them, staged UFLS 0.36, and the learned policy 0.03 with a recovery time of 2.3 s at a nadir of 59.8 Hz. The two results compose, because the forecaster identifies the hours that matter and the shed policy survives the ones it gets. A reliability surrogate and a maintenance scheduler were also built to complete the surrounding workflow, but they are separate screening-grade components and no claim above uses them.

5 Discussion and limitations

Standalone accuracy is never quoted as the result, because the forecaster's AUROC is evidence that a risk signal exists while the claim of value is the downstream blackout rate measured on exactly the hours the system itself selects. That distinction between a component that scores well and one that is safe inside the loop is the part that transfers to other safety-critical systems.

The limits are stated exactly. The frequency model is single-area and reduced-order rather than an AC power-flow model, so transmission constraints, line congestion, voltage collapse, and cascading outages are out of scope and no AC-feasibility claim is made. All data are synthetic, and real operator data would re-tune every number. The DQN is evaluated on the scenario family it was trained for, which is capacity-weighted unit failures at near-peak load, and the loop headline rests on 80 flagged hours over five test years with per-scenario outcomes saved for audit. That closed-loop rate is therefore a measurement over 80 events rather than a tight estimate of a rare-event probability, and we attach no confidence interval to it. The action set is ten discrete shed levels, matching the relay-adjacent practice this replaces, so no claim is made against continuous actor-critic baselines, which we did not run. Composition to multi-area transmission and safety certification of the shed policy are open directions rather than settled results.

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Availability of code and data

Every number is read from saved per-seed records and the trained checkpoint used by the closed loop, and the figure script asserts the values it plots, including the 0.382 mean blackout rate and the 0.03 closed-loop rate, so a stale figure fails loudly rather than shipping. The simulator, the four module implementations, the records and the checkpoint are available from the author on request, with environment pins and per-module reproduction commands. All data are generated by the included simulator and no proprietary data is used.

Use of AI tools

Generative AI tools were used for review and polish rather than for drafting: to audit the manuscript, to check every quantitative claim against the saved per-seed run records of the accompanying simulator, and to tighten the prose and citations against their sources. No model output was used as a research method inside the experiments. The authors directed the work and take responsibility for all content.